

Optimal Selection of Evolutionary based MPPT Algorithm for PV arrays under Multiple Partial Shading Conditions

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Abstract—Considering significant interest in clean energy in recent times, integration of photovoltaic (PV) systems has become in widespread. PV systems operating under partial shading conditions (PSCs) influences its profitability. Under PSC, PV system exhibit multiple local maximum power points (LMPPs) and one global maximum power point (GMPP). This work aims to study the performance and comparative analysis of 4 popular metaheuristic MPP algorithms under multiple partial shading conditions. The techniques studied are Flower Pollination Algorithm (FPA), Cuckoo Search Optimization Algorithm (CSO), Grey Wolf Optimization Algorithm (GWO), and Particle Swarm Optimization Algorithm (PSO). The tracking performance is evaluated under fast changing 5 shading scenarios. Results show that all the metaheuristic algorithms are found to have fast response and able to converge to their final steady state value within a short time of nearly 0.1s. However, for diverse shading scenarios, different algorithms are found to perform better. Thus a unique metaheuristic algorithm is not optimal for all shading scenarios. The energy yield calculations is proposed as a basis for optimal algorithm selection. GWO is identified as the optimal algorithm with the best energy output when shading scenarios have equal duration.

Index Terms—Dynamic Partial Shading, Flower Pollination Algorithm, Particle Swarm Optimization Algorithm, Grey Wolf Optimization Algorithm, Cuckoo Search Algorithm

I. INTRODUCTION

Despite significant cost reduction in PV Systems, energy harvesting optimization is still challenging. Although these systems include MPPT controllers, achieving MPPT becomes challenging due to partial shading conditions(PSC) [1]. In real-world scenarios, shading conditions can change rapidly due to moving clouds, dust accumulation, trees, buildings, objects, or other environmental factors. Partial shading results in hotspots, increase losses, and creates multiple peaks in the PV curve [2]. A partially shaded solar panel obstructs direct sunlight, which can significantly reduce energy efficiency and power output [3]. The issue of partial shading primarily affects photovoltaic (PV) modules, where the shading of even one cell can reduce the panel's overall efficiency. Partial shading can be mitigated using MPPT techniques, Multilevel inverters, bypass diodes, and array reconfiguration techniques. Reconfiguration techniques are suitable when there is a significant amount of shading, and the shading pattern is relatively constant.

Metaheuristic MPPT algorithms are suitable when shading is intermittent and the shading pattern is complex [5].

MPPT under dynamically changing partial shading conditions is challenging, as the PV array's power-voltage curve exhibits several maximas [4]. When traditional MPPT Algorithms such as Perturb and observe (P&O) and incremental conductance(INC) are employed to extract MPP under partially shaded conditions, it cannot differentiate between local and global MPP (LMPP and GMPP). It gets trapped in LMPP as it converges to the MPP which comes first which is most likely an LMPP. This leads to considerably low energy yield of up to 30%, significantly lower than the desired optimal GMPP [6]. Due to its success in managing PV array's non-linear behavior under PSCs, nature-inspired metaheuristic algorithms have recently gained popularity.

Metaheuristic algorithms can be applied to MPPT under dynamic shading scenarios to quickly and effectively find the optimal operating point of the PV system [7], [8]. These algorithms rely on real-time data from sensors that monitor solar irradiance to make informed decisions. As the solar irradiance varies, the solar power output also varies [9]. The variation in the solar power output indicates the variation in solar irradiance and shading patterns. If only one cell out of a solar panel of 60 cells gets shaded, it can drop the solar panel output by 30-40% [10]. Non-uniform solar irradiation leads to hotspot problems and can be reduced using a bypass diode, which helps pass the current from unshaded panels to shaded panels [11]. The efficient naturally inspired algorithms include the Particle Swarm Algorithm(PSO), Cuckoo Search Optimization Algorithm(CSO), Flower Pollination Algorithm(FPA), and Grey wolf Algorithm(GWO). Hybridization of these algorithms is done to improve the reliability and performance of the system [12].

In this work, the dynamic MPPT problem is formulated as an optimization problem aiming to maximize the PV system's power production under dynamically changing partial shading circumstances [13]. The decision variables to be optimized are the PV system's Voltage and Current set points. The paper aims to analyze a multiple shading system in different shading conditions [14].

II. SYSTEM OVERVIEW

The modelling and simulation of 5×4 solar PV array in Triple Tied (TT) configuration under dynamically changing environmental conditions is considered (Fig.1) [6]. Based on the Solar PV voltage and current the metaheuristic algorithm is used for Dynamic MPP Tracking which provides gate pulse to the DC-DC Boost converter. [15].

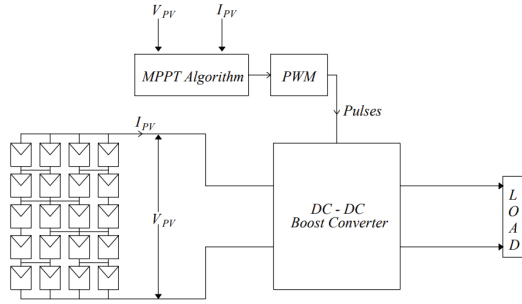


Fig. 1. Proposed 5 x 4 PV Array System in Triple Tied Configuration With MPPT Control Algorithm

The input data for the study includes solar PV specifications (Table I), ambient temperature, load profile, and irradiance profile. Irradiance profile includes uniform irradiance and multiple-shaded conditions. The electricity generated by the PV array depends on the Solar Irradiance, Ambient temperature, and the PV array terminal voltage.

TABLE I
PV MODULE SPECIFICATIONS

Parameters	Value
Make	HHV Technologies
Model	HSTUAF24295M
Cells per Module, N_{cell}	72
Rated power, P_{MPP}	295.16 W
Rated Voltage, V_{MPP}	36.85 V
Rated current, I_{MPP}	8.01 A
Open circuit Voltage, V_{oc}	44.6 V
Short circuit current, I_{sc}	8.49 A

The DC-DC boost converter can give a step-up voltage conversion by transforming lower voltage (V_{in}) to a higher voltage (V_{out}) [18]. The Boost Converter design parameters can be referred from Table II.

TABLE II
DESIGN PARAMETERS OF BOOST CONVERTER

Parameters	Value
Input Voltage, V_{in}	188.8 V
Output Voltage, V_{out}	241.5 V
Inductor, L	50 mH
Capacitor, C	2200 micro F
Resistor, R	10 ohms
Internal diode resistance, R_d	0.001 ohms
Forward Voltage, V_f	0.8 V
Switching frequency, F_s	25kHz

III. METHODOLOGY

The study explores the effectiveness of 4 meta-heuristic algorithms: Flower Pollination Algorithm (FPA) [16], Cuckoo Search Optimization Algorithm (CSO), Grey Wolf Optimization Algorithm (GWO) [17] and Particle Swarm Optimization Algorithm (PSO) under 5 partial shading scenarios in sequential multiple shading scenarios.

A. Mathematical modelling of PV

Several PV system models are used for partial shading studies, such as single diode, double diode, triple diode, and bypass diode models. The model's choice depends on the accuracy required for analysis. The single-diode model is popularly employed to simulate PV model behavior due to its balance between simplicity and accuracy. The model considers the effect of shading on each cell and combines the results to predict the overall system performance under partially shaded conditions. In this work, we use a single-diode model to simulate the I-V curve of each PV module (Fig.2).

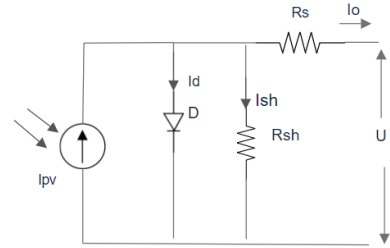


Fig. 2. Single diode Model

The PV module is modeled such that the Temperature and sun radiation influence its performance. The current generated by the PV module is calculated using [6] (1):

$$I_{pv} = I + I_o \left[\frac{U + IR_s}{e(nU_T)} - 1 \right] - \frac{U + TR_s}{R_{sh}} \quad (1)$$

where I is the load current (A), U is the PV module voltage, R_s is equivalent circuit series resistance in ohms, I_o is the reverse saturation current, U_T is the thermal voltage, n is the linearity factor (1 is for ideal diode) and R_{sh} is the ohms of the equivalent circuit shunt resistance.

B. Scenarios of Partial Shading

Depending on the solar irradiation on the PV modules, 5 different shading scenarios are applied to the PV arrangement. The shaded scenarios considered are: L-shaped, corner, center, bottom, and right side shading. 1000 W/m² of irradiance is considered for unshaded condition and 500 W/m² and 200 W/m² are considered for Partial Shaded conditions.

From the partial shading scenarios (Fig.3) the shading pattern for individual PV panels are obtained. For example, the insolation of the PV submodules PV11, PV12, PV21 and PV22 for the sequential shading pattern (SP) is computed in Table III.

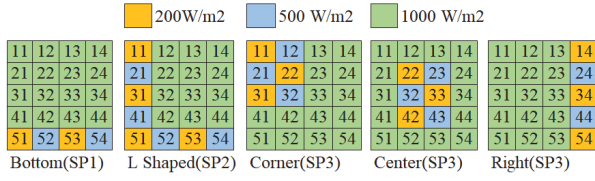


Fig. 3. Partial Shading scenarios and dynamic partial shading sequence

 TABLE III
PV PANEL IRRADIANCE (W/m²)-FOR DIVERSE SHADING PATTERNS

Pattern No.	PV11	PV12	PV21	PV22
SP1	1000	1000	1000	1000
SP2	200	1000	1000	1000
SP3	200	500	500	200
SP4	1000	1000	500	200
SP5	1000	1000	1000	1000

The value and Position of GMPP continuously changes with changes in shading scenarios. Thus the proposed MPPT algorithms should have the ability to track the GMPP whenever the shading conditions are changed. The dynamically changing complex shading pattern or sequence is defined next. The first 0.6s exhibits bottom shading, where the subsequent shading scenarios occurs in an interval of 0.6s in order : L-shaped, Corner, Center, and Right-Sided shading, respectively. In this context, an MPPT algorithm is more suitable when compared to PV array reconfiguration, as the shading pattern is intermittent and complex.

C. Objective Function

The objective function aims to maximize the PV system's output of power by tracking the MPP in real time, considering the effects of shading, temperature variations, and other factors that affect the efficiency of the PV system (2).

$$\text{Maximise } P_{\text{output}} = V_{pv} * I_{pv} \quad (2)$$

subject to

$$D_{\min} \leq D \leq D_{\max} \quad (3)$$

where, P_{output} represents the power output of the PV system, and V_{pv} and I_{pv} are the photovoltaic voltage and current, respectively; $D_{\min}$ and $D_{\max}$ are the minimum and maximum limits of the duty ratio (D).

D. Metaheuristic Algorithms

The Initialisation Stage is when the input parameters and control parameters of the MPPT algorithm are initialized [7]. In the duty cycle evaluation stage, the duty cycle is evaluated and the PWM signal to the DC-DC Power converter is successively generated. The best Duty cycle identification Stage is where the MPPT algorithm repeats the search process until the GMPP is obtained and the value of GMPP remains unchanged [14]. The weather condition monitoring stage is when the MPPT Algorithm continuously monitors the shading scenario to identify the changes in the weather conditions and reruns

the optimization when the change in the weather condition is observed.

1) *Flower Pollination Algorithm (FPA)*: The performance of each flower in the search space is assessed using the fitness function, and the algorithm then chooses the best flower out of all of them. Self-pollination or cross-pollination will be executed depending on the switch probability. In this algorithm, we use lavish flight or lavish distribution to determine the step size for the pollinators. No pollinators are necessary for local or self-pollination, and then we will evaluate the new solution. The best solution is then chosen after determining whether the new solution is preferable to the prior one or not. In the FPA method, each particle position is determined by (4):

$$x_i(t+1) = x_i^t + L(g^* - x_i^t) \quad (4)$$

The flowchart of FPA algorithm is summarized in Fig. 4 (a).

2) *Cuckoo Search Optimization (CSO)*: In CSO, the solution's fit is evaluated using the fitness function. We utilize the Levy flight equation/Levy distribution to determine how far the bird will fly from its current location to its destination. The initial stage of the algorithm's series of smaller phases is Levy Distribution. Levy Flight can then suddenly gain a larger step size, delivering a random walk. The New Solution is given by (5):

$$x_i(t+1) = x_i^t + (x_j^t - x_k^t) \quad (5)$$

where $i = 1, 2, 3, 4, \dots, n$ and t is the most latest iteration. The flowchart for CSO is summarized in Fig. 4 (b).

3) *Particle Swarm Optimization (PSO)*: The social behavior of fish and birds served as the basis for the computational optimization method known as particle swarm optimization (PSO) [17]. A popular technique for approximating solutions to challenging optimization and search problems is this population-based optimization method. PSO represents a potential optimization solution as a "particle" in a multidimensional search space. The position and velocity of each particle are updated across iterations to determine the best outcome [7]. PSO can be sensitive to parameter adjustments and may not always guarantee finding the global optimum [18]. The Particle position is Calculated using (6):

$$x_i(t+1) = x_i^t + v_i^t \quad (6)$$

The flowchart of PSO is shown in Fig. 4 (c)

4) *Grey Wolf Optimization (GWO)*: GWO is modeled based on the interpersonal hunting methods of a pack of wild grey wolves [18]. This method attempts to solve an optimization problem by simulating grey wolf's interpersonal interactions and hunting techniques. A population-based optimization algorithm strikes a balance between exploitation and exploration. It is based on the cooperation and social connections seen in the behavior of grey wolves in the wild [12]. The selection of the parameters and the features of the problem can affect performance, and parameter modification may be required for the best outcomes [14]. GWO is well-known for its

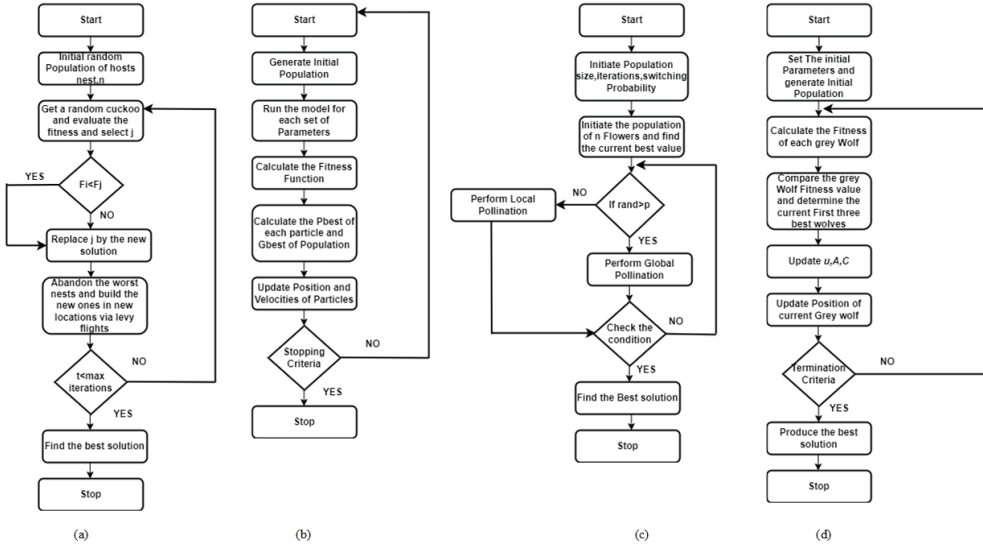


Fig. 4. Flowchart for (a)FPA (b)CSO (c)PSO and (d) GWO Algorithms

effectiveness in resolving various optimization issues. It has been used in several industries, including engineering, machine learning, and finance [13]. The updation of the positions of the alpha, beta and delta wolf are given by(7)-(8):

$$\vec{D} = |C \cdot \vec{X}_p(t) - \vec{X}_p(t)| \quad (7)$$

$$\vec{X}(t+1) = \vec{X}_p(t) - \vec{A} \cdot \vec{D} \quad (8)$$

where t is the current iteration, D, A, and C are the coefficient vectors, $\vec{X}_p$ and $\vec{X}$ are the position vectors of prey and grey wolf. The flowchart of GWO is shown in Fig. 4 (d).

E. Performance Indices

The effectiveness of a PV system under partial shadowing is evaluated using performance indices like Performance Ratio and mismatch losses. A mismatch between the solar cell's output currents in the PV module generates mismatch losses. The difference between the P_{mpp} of the array's ultimate I-V characteristics and the total P_{mpp} of each independent submodule is known as the mismatch loss (9):

$$MPL(\%) = \frac{P_{mpp} - P_{mpp}(PSC)}{P_{mpp}} \times 100 \quad (9)$$

where P_{mpp} is the maximum power generated during zero shading condition and $P_{mpp}(PSC)$ is the peak power obtained during partial shading scenarios

A metric used to evaluate the effectiveness and general performance of a solar PV (photovoltaic) system is the performance ratio (PR). This is calculated by (10)

$$PR = \frac{ActualPoweroutput}{Expectedpoweroutput} \times 100 \quad (10)$$

where *Actual Power Output* is the output power generated by the solar system and *Expected Power Output* is the power generated by the system under zero shading conditions

IV. RESULTS AND DISCUSSION

The 5×4 PV array TT configurations under zero shading condition, and dynamic Partial shading condition with MPPT is modeled in Matlab/Simulink R2022a. A technical comparison of 4 different meta-heuristic algorithms in TT configuration is conducted to find the best MPPT algorithm under multiple partial shading conditions.

A. Zero shading and Partial Shading Condition

Zero shading is when a solar panel is exposed to direct, unhindered sunlight. All PV modules are exposed to a constant temperature of 25°C and an uniform irradiation of 1000W/m². The system exhibits a single GMPP with a peak power of 5891 W and zero power loss on the P-V curve. The P-V Curve for Zero shading condition is given in Fig.5.

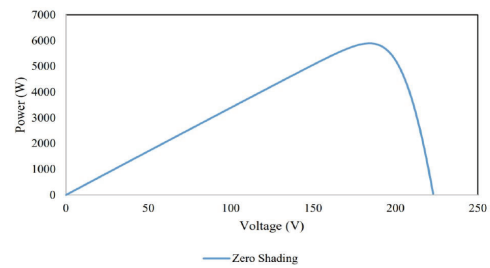


Fig. 5. P-V Curve - Zeroshading condition

The P-V curve under partial shading condition is given in Fig.6.

B. Multiple Shading scenarios

Considering the multiple shading scenarios, the tracking curves determine the dynamic responses of power(Fig.9), voltage (Fig.7), and current (Fig.8) of the PV system.

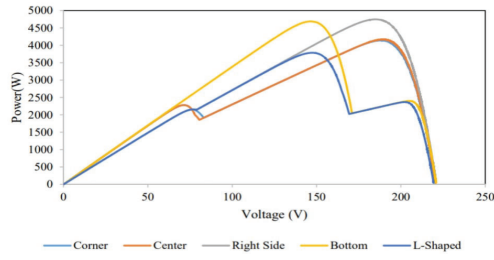


Fig. 6. P-V Curve - Partial shading condition

The voltage and current obtained are also compared for different algorithms. The maximum power obtained for each algorithm under multiple sequential shading scenario is noted. It is observed that all the Algorithms are converging to the final steady state values within a short time of nearly 0.1s. Thus Metaheuristic algorithms have fast response under complex and fast partial shading conditions. The MPP tracking results are summarised in Table IV and Fig. 10.

 TABLE IV
MAXIMUM POWER - MULTIPLE SHADING SCENARIOS

Sequence	P _{FPA} (W)	P _{CSO} (W)	P _{PSO} (W)	P _{GWO} (W)	MPL Best(%)	PR Best
Bottom	4725	4999	3447	3998	15.1	84.8
L Shaped	4137	4001	3120	3541	29.8	70.2
Corner	4356	4878	4155	4272	17.2	82.8
Center	3212	3657	4123	4161	29.4	70.6
Right side	4555	3545	4714	5123	13.0	86.9

For different shading scenarios, different algorithms perform better. Under bottom shading and corner shading, CSO performs better, whereas FPA performs better under L-Shaped shading. GWO performs better under corner and right side shading. A single and unique metaheuristic algorithm is not optimal for all shading scenarios. It becomes challenging to select an appropriate algorithm. Thus, we compute the total energy yield during the sequential multiple shading duration as in Table V.

 TABLE V
ENERGY YIELD-MULTIPLE SHADING SCENARIOS

Sequence	E _{FPA} (Ws)	E _{CSO} (Ws)	E _{PSO} (Ws)	E _{GWO} (Ws)	E _{Best} (Ws)	Best Case
Bottom	2835	2999.4	2068	2399	2999	CSO
L Shaped	2482	2401	1872	2125	2482	FPA
Corner	2614	2927	2493	2563	2927	CSO
Center	1927	2194	2474	2497	2497	GWO
Right side	2733	2127	2829	3074	3074	GWO
E _{total}	12591	12648	11735	12658	13979	GWO

The energy yield calculations show that GWO gives the best energy output (12657Ws in 3s). However, this inference is valid only when the shading scenarios have equal duration. Thus, depending on the duration of shading scenarios in a geographical location based on energy yield calculation, the optimal algorithm has to be selected. However, if the best

algorithm can be identified and implemented in each shading scenario, the energy yield can be further improved to an estimated 13978.8 kWh in 3s.

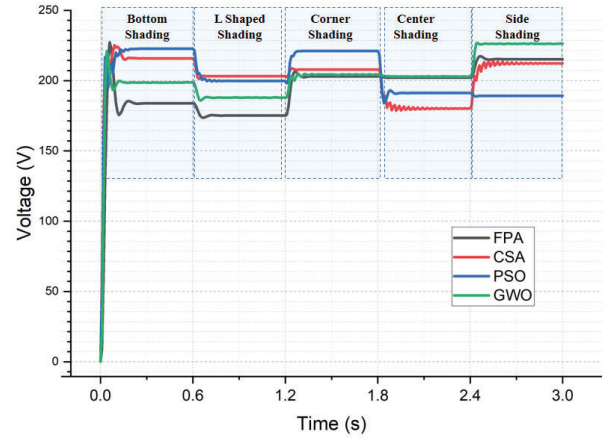


Fig. 7. Voltage-Time characteristics- Multiple shading scenarios

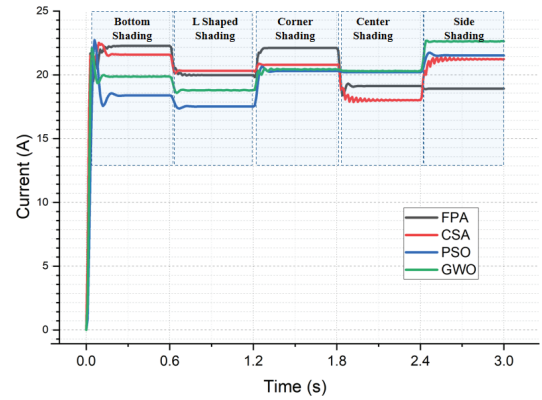


Fig. 8. Current-Time characteristics- Multiple shading scenarios

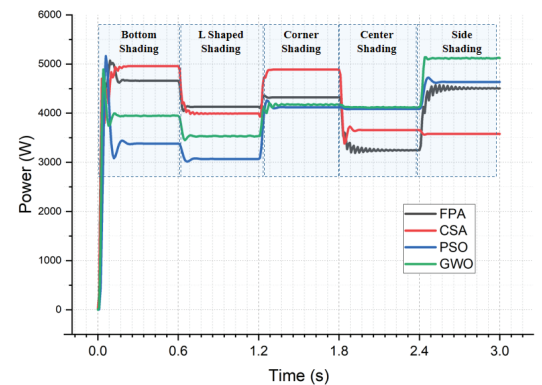


Fig. 9. Power-Time characteristics- Multiple shading scenarios

V. CONCLUSION

MPPT Tracking under complex and dynamically changing partial shading conditions requires the proper control algo-



Fig. 10. Multiple shading scenario MPP Tracking

rithms with fast response time and the ability to adapt to rapidly changing environmental conditions. The research explores the effectiveness of 4 meta-heuristic algorithms: Flower Pollination Algorithm (FPA), Cuckoo Search Optimization Algorithm (CSO), Grey Wolf Optimization Algorithm (GWO), and Particle Swarm Optimization Algorithm (PSO) under 5 partial shading scenarios in independent mode as well as sequential mode. The Triple-Tied PV configuration is considered because it provides better reconfiguration to reduce the mismatch penalty. The Study result shows that for different shading scenarios, different algorithms perform better. We compute the total energy yield during the sequential multiple shading duration to select the appropriate algorithm. The energy yield calculations show that GWO gives the best energy output when shading scenarios have equal duration. Thus, using metaheuristic algorithms for MPPT under dynamically changing partial shading conditions offers the advantage of adaptability and the ability to explore complex solution spaces. The energy yield can be improved if the best algorithm can be identified and implemented in each shading scenario. However, implementing such algorithms is computationally intensive and the results obtained by metaheuristic algorithms have randomness. Properly tuned and integrated algorithms can significantly improve PV System's efficiency and energy yield in dynamic and challenging environments.

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